



Department Of Computer Science, CUI Lahore Campus

CSC102 - Discrete Structures

Course Outline

- Nested Quantifiers

Nested Quantifiers

- Two quantifiers are nested if one is within scope of other, such as

$$\forall x \exists y (x + y = 0).$$

- Everything within the scope of a quantifier can be thought of as a propositional function.
- For example,
 $\forall x \exists y (x + y = 0)$
is the same thing as $\forall x Q(x)$, where $Q(x)$ is $\exists y P(x, y)$, where $P(x, y)$ is $x + y = 0$.

Nested Quantifiers

- $\forall x \exists y P(x, y)$
 - “**For all** x , there exists a y such that $P(x, y)$ ”.
 - Example:
 - $\forall x \exists y (x + y = 0)$ where x and y are integers
- $\exists x \forall y P(x, y)$
 - **There exists an** x such that for all y , $P(x, y)$ is true”
 - Example: $\exists x \forall y (x \times y = 0)$
- $\forall x \forall y \forall z (x + (y + z) = (x + y) + z)$
- **THINK QUANTIFICATION AS LOOPS**

Nested Quantifiers Example

- Let Domain of x is the students in this class
Domain of y is the courses in 2nd semester of BS(CS)
 $Q(x, y)$ = “ x takes course y ”, true when x takes course y , otherwise false.

Translate the following logical expression:

- $\forall x \forall y Q(x, y)$
- $\exists x \exists y Q(x, y)$
- $\forall x \exists y Q(x, y)$
- $\exists x \forall y Q(x, y)$

Meaning of Multiple Quantifiers

Suppose $P(x, y) = \text{"x likes y."}$

Domain of x : {St1, St2}; Domain of y : {Cricket, Hockey}

- $\forall x \forall y P(x, y)$
 - $P(x, y)$ true for all x, y pairs.
- $\exists x \exists y P(x, y)$
 - $P(x, y)$ true for at least one x, y pair.
- $\forall x \exists y P(x, y)$
 - For every value of x we can find a (possibly different) y so that $P(x, y)$ is true.
- $\exists x \forall y P(x, y)$
 - There is at least one x for which $P(x, y)$ is always true.

Predicates - the meaning of multiple quantifiers

<i>Statement</i>	<i>When True?</i>	<i>When False?</i>
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y .

Example

- Let $Q(x, y)$: “ $x + y = 0$ ”

What are the truth values of the quantifications $\exists y \forall x Q(x, y)$ and $\forall x \exists y Q(x, y)$, where the domain for all variables consists of all real numbers?

Example

- Let $Q(x, y)$ denote “ $x + y = 0$.” What are the truth values of the quantifications $\exists y \forall x Q(x, y)$ and $\forall x \exists y Q(x, y)$, where the domain for all variables consists of all real numbers?

- Solution:**

The quantification $\exists y \forall x Q(x, y)$ denotes the proposition

“There is a real number y such that for every real number x , $Q(x, y)$.”

- No matter what value of y is chosen, there is only one value of x for which $x + y = 0$. Because there is no real number y such that $x + y = 0$ for all real numbers x , the statement $\exists y \forall x Q(x, y)$ is false.

Example

- The quantification $\forall x \exists y Q(x, y)$ denotes the proposition
“For every real number x there is a real number y such that $Q(x, y)$.”
- Given a real number x , there is a real number y such that $x + y = 0$; namely, $y = -x$.
- Hence, the statement $\forall x \exists y Q(x, y)$ is true.

Order of Quantifiers

- $\exists x \forall y$ and $\forall x \exists y$ are not equivalent!
- $\exists x \forall y P(x,y)$
 - $P(x,y) = (x+y == 0)$ is false
- $\forall x \exists y P(x,y)$
 - $P(x,y) = (x+y == 0)$ is true

Example

$Q(x, y, z): x + y = z$

Domain: Real numbers

- $\forall x \forall y \exists z Q(x, y, z)$ True/False???
- For all real numbers x and for all real numbers y there is a real number z such that $x + y = z$.
- **True**
- $\exists z \forall x \forall y Q(x, y, z)$ True/False???
- There is a real number z such that for all real numbers x and for all real numbers y that $x + y = z$.
- **False**

Negating Multiple Quantifiers

- Recall negation rules for single quantifiers:
 - $\neg \forall x P(x) = \exists x \neg P(x)$
 - $\neg \exists x P(x) = \forall x \neg P(x)$
 - Essentially, you change the quantifier(s), and negate what it's quantifying
- Examples:
 - $\neg(\forall x \exists y P(x,y)) = \exists x \neg \exists y P(x,y) = \exists x \forall y \neg P(x,y)$
 - $\neg(\forall x \exists y \forall z P(x,y,z)) = \exists x \neg \exists y \forall z P(x,y,z)$
 $= \exists x \forall y \neg \forall z P(x,y,z) = \exists x \forall y \exists z \neg P(x,y,z)$

Negating Multiple Quantifiers

- Consider $\neg(\forall x \exists y P(x,y)) = \exists x \forall y \neg P(x,y)$
 - The left side is saying “for all x , there exists a y such that P is true”
 - To negate it, you need to show that “there exists an x such that for all y , P is false”
- Consider $\neg(\exists x \forall y P(x,y)) = \forall x \exists y \neg P(x,y)$
 - The left side is saying “there exists an x such that for all y , P is true”
 - To negate it, you need to show that “for all x , there exists a y such that P is false”

Chapter Reading

- **Chapter 1**, Kenneth H. Rosen, Discrete Mathematics and Its Applications, Section 1.5

Chapter Exercise (For Practice)

- Question # 1, 2, 3, 4, 24, 25, 26, 27, 30, 31